By the second fundamental theorem of integral calculus, the following properties of definite integrals hold. They are stated here without proof.

Property 1 : $\int_a^b f(x)\,dx = \int_a^b f(u)\,du$, $a < b$

i.e., definite integral is independent of the change of variable.

Property 2 : $\int_a^b f(x)dx = -\int_b^a f(x)dx$

i.e., the value of the definite integral changes by minus sign if the limits are interchanged.

Property 3 : $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$, $a < c < b$

Property 4 : $\int_a^b [\alpha f(x) + \beta g(x)]dx = \alpha\int_a^b f(x)dx + \beta\int_a^b g(x)dx$, where α and β are constants.

Property 5 : If $x = g(u)$, then $\int_a^b f(x)dx = \int_c^d f(g(u))\frac{dg(u)}{du}du$ where $g(c) = a$ and $g(d) = b$.

This property is used for evaluating definite integrals by making substitution.

We illustrate the use of the above properties by the following examples.

Example 9.5

Evaluate : $\int_0^3 (3x^2 - 4x + 5)dx$.

Solution

$$\begin{aligned}\int_0^3 (3x^2 - 4x + 5)dx &= \int_0^3 3x^2dx - \int_0^3 4x\ dx + \int_0^3 5\,dx \\ &= 3\int_0^3 x^2dx - 4\int_0^3 x\,dx + 5\int_0^3 dx \\ &= 3\left[\frac{x^3}{3}\right]_0^3 - 4\left[\frac{x^2}{2}\right]_0^3 + 5[x]_0^3 \\ &= (27-0) - 2(9-0) + 5(3-0) \\ &= 27 - 18 + 15 = 24\,.\end{aligned}$$

Example 9.6

Evaluate : $\int_0^1 \frac{2x+7}{5x^2+9}dx$.

Solution

$$\int_0^1 \frac{2x+7}{5x^2+9}dx = \int_0^1 \frac{2x}{5x^2+9} + 7\int_0^1 \frac{dx}{(5x^2)+3^2} = \frac{1}{5}\log[5x^2+9]_0^1 + \frac{7}{5}\int_0^1 \frac{dx}{x^2 + \left(\frac{3}{\sqrt{5}}\right)^2}$$

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